Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, R, SQL, Java, C/C++
Statistical & Quantitative Analysis: Predictive Modeling, Probabilistic Modeling (BG/NBD, Gamma-Gamma), Time-Series Analysis, Anomaly Detection, Feature Engineering, Sentiment Analysis (FinBERT)
Data Science Tools: Pandas, NumPy, Scikit-learn, Power BI, Streamlit, Tableau
Software Engineering: Data Structures & Algorithms, Operating Systems, Database Management Systems (DBMS), Multithreading & Concurrency, Unit Testing & TDD
Databases: MySQL, SQLite, Redis
Tools & Platforms: Git, GitHub, Jupyter Notebook

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** — ***Aug – Sep, 2025

- Analyzed 8,500+ real-world records using Julius AI and LLM-assisted workflows, extracting 15 actionable business insights through data exploration and prompt engineering.
- Developed structured analytical workflows, validated AI-generated outputs, and translated findings into 4 business recommendations presented to 3 mentors.
- Completed an industry-sponsored capstone project, earning certification from VOIS, Vodafone Idea Foundation, and AICTE.

PROJECTS

***Real-Time Stock Anomaly Detector* |LINK**

- Conducted end-to-end quantitative analysis on 2 years of NSE stock market data, engineering 9 financial features (RSI, Bollinger Bands, volume ratio) to model market behavior statistically.
- Built and evaluated dual predictive models — Isolation Forest and LSTM Autoencoder — detecting anomalous trading patterns across 448 sessions, with model outputs validated against real market events.
- Integrated FinBERT sentiment analysis on live news headlines to combine quantitative price signals with qualitative market sentiment, delivered through an interactive dashboard for risk-flagging.

***Customer Lifetime Value Engine* |LINK**

- Developed probabilistic forecasting models (BG/NBD, Gamma-Gamma) on 1M+ retail transactions to predict 12-month Customer Lifetime Value for 4,000+ customers.
- Built churn-risk scoring and value-based segmentation to quantify customer risk and inform data-driven retention strategy.
- Delivered findings through a Power BI dashboard and a Streamlit-based ROI simulator, translating statistical outputs into measurable business recommendations.

***RedisLite* |LINK**

- Engineered a Redis-inspired in-memory key-value store in Python with a custom RESP protocol parser and multithreaded TCP server, demonstrating strong algorithmic and systems-level programming.
- Implemented Write-Ahead Logging and crash recovery, validating correctness with 46 automated tests and benchmarking 619K SET ops/sec.

***API Rate Limiter Library* |LINK**

- Built a production-ready Python API rate-limiting library with five industry-standard throttling algorithms, Redis integration, Flask/FastAPI middleware, thread-safe request handling, configurable storage backends, and PyPI distribution.
- Developed and validated a modular, reusable architecture with 30+ PyTest unit tests using TDD practices, covering edge cases, concurrency behavior, and algorithm correctness to ensure production-grade reliability.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate |**LINK** 2026
- Deloitte Australia Data Analytics Job Simulation |**LINK** 2025